SECTION 1

Time—30 minutes

45 QUESTIONS

r each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

ch question below consists of a word in capital letters, llowed by five lettered words or phrases. Choose the rd or phrase that is most nearly opposite in meaning the word in capital letters. Since some of the ques- ns require you to distinguish fine shades of meaning, nsider all the choices before deciding which is best.

ample:

OOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

. FRET: (A) explain (B) watch (C) relax (D) plan (E) invite

. DISHEARTEN: (A) enroll (B) mislead (C) distort (D) prepare (E) encourage

. LUXURIANT: (A) scanty (B) unskilled (C) reverent (D) shameful (E) unpredicted

. HINDRANCE: (A) expression (B) preference (C) assistance (D) enjoyment (E) emphasis

. INANIMATE: (A) somber (B) valiant (C) supportive (D) hidden (E) alive

. ANONYMOUS: (A) synonymous (B) signed (C) modern (D) proven (E) temporary

. ATYPICAL: (A) favorable (B) emotional (C) factual (D) usual (E) moral

. CONCEIT: (A) self-effacement (B) self-sufficiency (C) ill repute (D) ill humor (E) unreceptiveness

. RANT: (A) hold together (B) happen seldom (C) work diligently (D) speak softly (E) step lightly

. WOODEN: (A) lucky (B) haughty (C) trustworthy (D) graceful (E) proper

. ACQUIESCE: (A) combine (B) gain (C) panic (D) protect (E) object

. ADROIT: (A) impure (B) harmless (C) secretive (D) abstract (E) inept

. MUNIFICENT: (A) judicious (B) obedient (C) miserly (D) determined (E) suggestive

. UNDERSTATE: (A) placate (B) insinuate (C) embroider (D) dissemble (E) rebound

. VIRULENT: (A) sectarian (B) defensible (C) comparable (D) cowardly (E) innocuous

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ---, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. Historical buildings in many American cities, rather than being destroyed, are now being ----.

(A) constructed (B) condemned (C) described (D) renovated (E) designed

17. Sadly, many tropical rain forests are so ---- by agricultural and industrial overdevelopment that they may ---- by the end of the century.

(A) isolated. .separate
(B) threatened. .vanish
(C) consumed. .expand
(D) augmented. .diminish
(E) rejuvenated. .disappear

18. Like all reformers, Wood is ---- who believes that the world can be changed by good people promoting right deeds.

(A) a statistician (B) a prodigal (C) a legislator (D) an opportunist (E) an optimist

19. The ---- of the Navajos is reflected in the fact that many have retained their language and ancient customs.

(A) diversity (B) adaptability (C) modernization (D) cohesiveness (E) creativity

20. I am astounded that the author of such heretofore --- articles could generate such spritely prose and maintain such an original narrative.

(A) memorable (B) pedestrian (C) perceptive (D) enthralling (E) effervescent

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1

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

(This passage was written in 1959.)

Although at least one school of thought supports the thesis that the people of African descent in the Western world have no African heritage to reclaim, I am not of that school. Many writers, both Black and White, have pointed to a rich and ancient heritage which, in my opinion, we must affirm resolutely if Black Americans in general and Black writers in particular are ever to be reconciled with their roots.

As far back as 1881, the renowned scholar, Dr. Edward Blyden, sounded the note for the organized teaching of African culture and civilization. In 1883 George W. Williams, the first Black member of the Ohio legislature and the person responsible for the first organized African studies in the United States, devoted 125 pages of his *History of the Negro Race in America* to background information about African culture and society. The field of research in African history was widened later under the leadership of three men in particular: Dr. W. E. B. Du Bois, Dr. Jesse E. Moorland, and Dr. Carter G. Woodson. Black America's mission to reclaim its lost African heritage was well underway.

What is this heritage? In the first place, the history, art, and folklore of West Africa, the ancestral home of most Black Americans, offer proof that Black people built great nations and an advanced culture hundreds of years before their first appearance in Jamestown, Virginia, in 1619. During the period when Timbuktu was the great intellectual center of the Songhai empire, African scholars enjoyed a renaissance known and respected throughout most of Africa and parts of Europe. Further, long before the European colonial period, independent nations were already established in Africa; many of them possessed long and glorious histories. For example, during the reign of Askia the Great, who came to power a year after Columbus discovered America, the nations of West Africa enjoyed a standard of life equal to, and often higher than, that of other nations of the world.

Historically, Africa and its peoples have been viewed mainly through European eyes and measured by European standards. The entire history of Africa must be rewritten in a manner that challenges and reverses long-held European concepts. This task is singularly the responsibility of Black writers. An abundance of material on African life awaits their attention. They would do well to consider using as a guide two sentences from John W. Vandercook's book *Tom-Tom*.

> A race is like a man. Until it uses its own talents, takes pride in its own history, and loves its own memories, it can never fulfill itself completely.

The reading passages in this test are brief excerpts or adaptations of excerpts from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

21. With which of the following statements regarding African culture would the author most likely agree?
    (A) Africans had highly developed cultures long before their first contact with Europeans.
    (B) African culture will not flourish again until Black Americans reclaim it.
    (C) The work of European historians provides a sound basis for further research in African cultural history.
    (D) The scarcity of material on African life has hindered the efforts of Black Americans to reclaim their heritage.
    (E) The cultural significance of the art, history, and folklore of Africa was not universally acknowledged until the 1880's.

22. The author cites Du Bois, Moorland, and Woodson for their
    (A) strong community leadership
    (B) political achievements
    (C) teaching activities
    (D) scholarly investigations of African history
    (E) benevolence toward West Africa

23. The author suggests that European accounts of African history have been
    (A) culturally biased
    (B) avidly read
    (C) systematically suppressed
    (D) adequately refuted
    (E) extensively researched

24. It can be inferred that the author expects Vandercook's quoted statement to be regarded as
    (A) evidence of the rich and ancient heritage of Black Americans
    (B) consistent with European attitudes regarding African history
    (C) a source of inspiration for Black historians and writers
    (D) a stylistic model that will be imitated by fledgling writers
    (E) a tribute to the literary accomplishments of Black writers to date

25. Which of the following best describes the purpose of the passage?
    (A) To describe the influence of African culture on the Western world
    (B) To defend the validity of African studies programs in United States universities
    (C) To encourage Americans of African descent to reclaim their ancestry
    (D) To investigate the origin of the European concept of African history
    (E) To outline appropriate methods for the documentation of Africa's cultural development

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Three main forms of epidemiological investigation are descriptive, analytical, and experimental. Descriptive epidemiology describes events in terms of the persons affected, the place at which the event occurred, and the time frame under consideration. An example of how a descriptive study shows the distribution of disease is the routine surveillance of infections in a hospital. Descriptive epidemiological data obtained from surveillance of hospital-acquired infection may provide information for early recognition of problems, either by detecting an excessive number of cases or, as on several occasions, by identifying the presence of a rare microorganism.

To find the cause of a disease or other problem, investigators follow the techniques of one of the two general types of analytic epidemiological study: cohort and case-control. A cohort study examines two groups—those with and those without an attribute suspected to have some relation to the disease. None of the individuals is a "case" at the outset. A cohort study determines whether or not the group displaying a certain characteristic develops the disease more often than does the group without the characteristic.

A case-control study asks whether the people who have the disease display certain characteristics more frequently than do those who are not affected. "Cases" are defined as those individuals who already have the disease. "Controls," used for comparison, are individuals without the disease. A case-control study is retrospective. The disease or condition has already developed, and researchers look for factors that preceded it. Retrospective studies are faster, less expensive, and better suited to analysis of rare diseases than are prospective studies, which follow a disease-free cohort over time, awaiting the occurrence of the disease. Since not all those exposed to it will develop the disease, large numbers of persons must be included in the prospective study population.

In experimental epidemiology the investigator deliberately manipulates a population, exposing part to the proposed risk or benefit factor and leaving another part unexposed. Risk or benefit factors are those that are thought to be causally related to a condition. (Epidemiologists often prefer the term "condition" to "disease" because an epidemiologist, in contrast to a physician or therapist, focuses interest on subclinical conditions and even on healthy members of the group under study.) Experimental studies are not difficult to justify when the intervention is relatively safe or when the patient is already at risk. However, performing such experimental studies is unacceptable in certain circumstances.

26. The main purpose of the passage is to

(A) compare and contrast different statistical theories
(B) explain common epidemiological research methods
(C) provide historical background on epidemiology
(D) describe specific infection-control practices
(E) emphasize the importance of sound research techniques

27. It can be inferred from the passage that descriptive epidemiology allows an investigator to answer routinely which of the following questions?

(A) What differences exist between the experimental and control groups?
(B) What steps are being taken to control the disease?
(C) How many people in a particular area have the disease?
(D) What specific factors prevented certain people from contracting the disease?
(E) How much information is needed to describe future effects of the disease?

28. According to the passage, a cohort study is one that

(A) gathers retrospective data
(B) is based at its outset on persons who have not yet contracted a given disease
(C) involves treatment of populations
(D) compares those who already have a given disease with those who do not
(E) is an example of descriptive epidemiology

29. The passage implies that an experimental epidemiological study is most likely to focus on a comparison between which of the following?

(A) The number of people in an untreated population who recover from a disease and the number who do not
(B) The occurrence of a disease among those who have and those who do not have a given attribute
(C) The number of people over a certain age with a disease and the number under that age without the disease
(D) The infection rates after treating part of a group with one procedure and the other part with another procedure
(E) The patient population in one area and that in another area

30. According to the passage, an epidemiologist would consider an attribute to be a risk factor if it were thought to be

(A) a result of a disease
(B) a cause of a disease
(C) a symptom of a disease
(D) indicative of an incurable disease
(E) unrelated to the disease under study

GO ON TO THE NEXT PAGE

1

Select the word or set of words that best completes each of the following sentences.

31. Hoping to ---- the angry customer, the manager of the store courteously offered a refund on the purchase.

(A) pique (B) upbraid (C) embolden
(D) conciliate (E) compliment

32. Even those who ---- the aphorism "history repeats itself" must now ---- the fact that certain events bear marked resemblances to incidents of earlier history.

(A) reject. .doubt
(B) deny. .acknowledge
(C) defend. .perceive
(D) ignore. .forget
(E) heed. .admit

33. Scientific research has become so ---- that even leading scientists are now expected to ---- their work in order to obtain grants.

(A) expensive. .justify (B) vital. .propose
(C) popular. .belittle (D) exact. .minimize
(E) esoteric. .falsify

34. My plea is not for drab and ---- technical writing about music but for pertinent information conveyed with as much ---- as possible.

(A) repetitive. .redundancy
(B) obscure. .felicity
(C) inscrutable. .ambivalence
(D) euphonious. .harmony
(E) provocative. .exhilaration

35. Ms. Warner supplements her selection of Queen Victoria's drawings with a text as idiosyncratic as it is ----, for she can be ---- as well as a scholarly writer.

(A) professional. .a rigorous
(B) perplexing. .a popular
(C) unique. .a conscientious
(D) entertaining. .a traditional
(E) accurate. .an unconventional

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. INTRODUCTION : CONCLUSION ::
(A) announcement : news
(B) greeting : farewell
(C) birth : marriage
(D) arrival : salutation
(E) friendship : termination

37. STRIKE : WORK :: (A) postpone : cancel
(B) censor : learn (C) incorporate : merge
(D) detour : trespass (E) boycott : purchase

38. SCALPEL : SURGEON :: (A) razor : barber
(B) weed : gardener (C) recipe : chef
(D) medicine : patient (E) compass : engineer

39. PREPOSTEROUS : COMMON SENSE ::
(A) illegal : law (B) chronic : pain
(C) indelible : error (D) justifiable : logic
(E) hilarious : laughter

40. CLASH : COLOR :: (A) intensity : sensation
(B) glimpse : vision (C) pebble : texture
(D) dissonance : sound (E) absurdity : perception

41. FORMIDABLE : FEAR :: (A) intolerable : patience
(B) grateful : regret (C) sickening : disgust
(D) generous : pity (E) dismal : cheer

42. BASTION : DEFENSE ::
(A) arsenal : storage
(B) anchorage : supply
(C) citadel : concealment
(D) asylum : embarkation
(E) pavilion : fortification

43. SAGA : LENGTHY :: (A) proverb : pithy
(B) eulogy : candid (C) poem : humorous
(D) play : short-lived (E) novel : acclaimed

44. BIRDS : AVIARY :: (A) bees : garden
(B) cows : herd (C) apes : jungle
(D) quails : bevy (E) sheep : fold

45. LASSITUDE : VIGOR ::
(A) dedication : employment
(B) greed : possessions
(C) indigence : funds
(D) rehearsal : performance
(E) repudiation : theory

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 2

Time—30 minutes

25 QUESTIONS

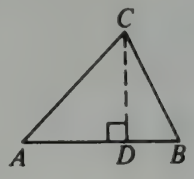
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \dfrac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. If $\frac{1}{N} + \frac{1}{N} = 6$, then $N =$

   (A) $\frac{1}{6}$

   (B) $\frac{1}{3}$

   (C) $\frac{1}{2}$

   (D) 2

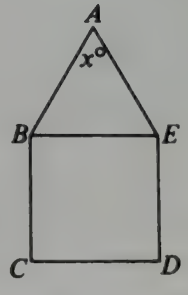
   (E) 3

2. If $x = 1$ and $y = -2$, then $2x - 3y =$

   (A) −7 (B) −4 (C) −1 (D) 5 (E) 8

3. In a senior class there are 200 boys and 300 girls. If 40 percent of the senior boys and 50 percent of the senior girls bought class rings, how many seniors bought class rings?

   (A) 200
   (B) 225
   (C) 230
   (D) 250
   (E) 275



4. If, in the figure above, $BCDE$ is a square and $AB = AE = DE$, then $x =$

   (A) 15 (B) 30 (C) 45 (D) 60 (E) 90

---

5. Last year Mr. Brown received 5 percent interest on an investment of $32,000. How much interest did he receive on this investment?

   (A) $160 (B) $640 (C) $1,600
   (D) $6,400 (E) $16,000

6. The sum, product, and average (arithmetic mean) of three integers are equal. If two of the integers are 0 and −5, the third integer is

   (A) −5
   (B) 0
   (C) 2
   (D) 5
   (E) 10

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2

Questions 7-9 refer to the following definition.

A	*B*
C	*D*

A block sum is a figure, like the one above, that has the following properties:

1. $A, B, C,$ and D are digits from 1 to 9 inclusive.
2. $A + B = 10C + D$.

For example, | 5 | 8 | / | 1 | 3 | is a block sum because 1, 3, 5, and 8 satisfy property 1 and $5 + 8 = 13$ satisfies property 2.

A	*B*
1	2

7. If the figure above is a block sum, what is the value of $A + B$?

   (A) 12 (B) 3 (C) 2 (D) 1
   (E) It cannot be determined from the information given.

9	7
C	*D*

8. If the figure above is a block sum, what digit does D represent?

   (A) 1 (B) 3 (C) 6 (D) 16
   (E) It cannot be determined from the information given.

A	*B*
1	1

9. If the figure above is a block sum, what is the value of A?

   (A) 1 (B) 2 (C) 9 (D) 11
   (E) It cannot be determined from the information given.

10. $90 + 6 + \frac{2}{5} + \frac{3}{50} =$

    (A) 90.623
    (B) 90.646
    (C) 96.046
    (D) 96.23
    (E) 96.46

11. A mechanic can install carburetors in 3 cars every 4 hours. At that rate, how long will it take the mechanic to install carburetors in 5 cars?

    (A) 6 hr. 20 min.
    (B) 6 hr. 40 min.
    (C) 7 hr. 15 min.
    (D) 7 hr. 30 min.
    (E) 7 hr. 45 min.

12. If Joe is now x years old and Marie is 6 years older than Joe, which of the following gives Marie's age 8 years ago?

    (A) $x - 14$ (B) $x - 6$ (C) $x - 2$
    (D) $6x - 8$ (E) $x + 14$

13. In flying a plane, a standard rate turn is one in which the rate of turning is 3 degrees per second. A heading is the number of degrees in a clockwise direction from north to the path of the plane. If a plane has a heading of 246° and makes a standard rate turn to a heading of 171°, what is the least number of seconds required for the turn?

    (A) 15
    (B) 25
    (C) 57
    (D) 82
    (E) 95

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14. Every student who studies art in a certain school receives exactly one of the grades A, B, C, or D. If $\frac{1}{5}$ of the students receive A's, $\frac{1}{4}$ receive B's, $\frac{1}{2}$ receive C's, and 10 students receive D's, how many students in the school study art?

(A) 50
(B) 60
(C) 90
(D) 100
(E) 200

P	Q	R
2	3	6
4	6	12
6	9	18
8	12	24
10	15	30
12	18	36

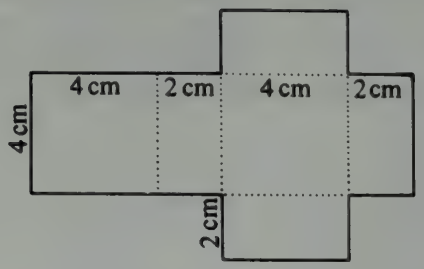
15. In the figure above, if p, q, and r are numbers from columns P, Q, and R respectively, then the greatest possible value of $\frac{p+r}{q}$ is

(A) $2\frac{2}{3}$ (B) $10\frac{2}{3}$ (C) 16 (D) 18 (E) 22

---

16. If $a \neq 0$, then $\frac{(-2a)^3}{-2a^3} =$

(A) -4 (B) -1 (C) 1 (D) 3 (E) 4



17. A closed rectangular box is formed by folding the figure above along the dotted lines. What is the volume of the box in cubic centimeters?

(A) 8 (B) 16 (C) 32 (D) 64 (E) 128

---

18. In a certain sequence of numbers, each number except the first is 5 plus twice the previous number. If the third number is 7, what is the first number?

(A) $-\frac{23}{4}$ (B) $-\frac{7}{2}$ (C) -2 (D) 1 (E) 43

19. If $y = 3x$, then the average (arithmetic mean) of x and y, in terms of x, is equal to

(A) x (B) $2x$ (C) $3x$ (D) $4x$ (E) $6x$

20. In a certain lawn-mower factory, 0.06 percent of all mowers produced are defective. On the average, there will be 3 defective mowers out of how many produced?

(A) 5,000
(B) 2,000
(C) 500
(D) 200
(E) 20

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21. If a square and an isosceles right triangle have equal areas, then the ratio $\frac{\text{length of leg of the triangle}}{\text{length of side of the square}} =$

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) $\frac{\sqrt{2}}{2}$

(D) $\frac{1}{1}$

(E) $\frac{\sqrt{2}}{1}$

22. If $a = 2d + 3$ and $b = 4d^2$, what is b in terms of a?

(A) $(a - 3)^2$

(B) $2(a - 3)^2$

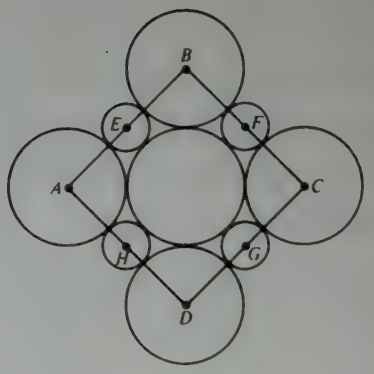
(C) $\frac{(a - 3)^2}{4}$

(D) $\frac{(a + 3)^2}{4}$

(E) $a + 3$

23. If the length and width of rectangle A are 10 percent less and 30 percent less, respectively, than the length and width of rectangle B, the area of A is equal to what percent of the area of B?

(A) 63%
(B) 60%
(C) 40%
(D) 6%
(E) 3%



24. Four circles of radius 2 with centers A, B, C, and D are arranged symmetrically around another circle of radius 2, and four smaller equal circles with centers E, F, G, and H each touch three of the larger circles as shown in the figure above. What is the radius of one of the small circles?

(A) $\sqrt{2} - 2$ (B) $\sqrt{2} - 1$ (C) $2\sqrt{2} - 2$
(D) 1 (E) $3\sqrt{2} - 1$

25. To make an orange dye, 3 parts of red dye are mixed with 2 parts of yellow dye. To make a green dye, 2 parts of blue dye are mixed with 1 part of yellow dye. If equal amounts of green and orange are mixed, what is the proportion of yellow dye in the new mixture?

(A) $\frac{3}{16}$

(B) $\frac{1}{4}$

(C) $\frac{11}{30}$

(D) $\frac{3}{8}$

(E) $\frac{7}{12}$

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 3

Time—30 minutes

40 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. DETERIORATION: (A) haste (B) diversity (C) evasion (D) development (E) concealment

2. DRASTIC: (A) inconsistent (B) accidental (C) creative (D) familiar and pleasing (E) mild and ineffective

3. FORMALITY: (A) lack of variation (B) lack of ceremony (C) lack of emphasis (D) solitude (E) insignificance

4. ASTRONOMICAL: (A) unclouded (B) occasional (C) very small (D) outwardly calm (E) approximately equal

5. BUSTLE: (A) forget (B) dawdle (C) protect (D) correct (E) garble

6. PRODIGY: (A) proclamation (B) maneuver (C) vague suggestion (D) ordinary occurrence (E) foolish explanation

7. STOKE: (A) deny (B) release (C) smother (D) tighten (E) stiffen

8. SUBLIMINAL: (A) debased (B) superlative (C) easily excited (D) consciously perceived (E) brightly colored

9. POTABLE: (A) unable to react (B) unfit to drink (C) irreplaceable (D) agreeable (E) immeasurable

10. EFFRONTERY: (A) need for change (B) show of solidarity (C) diffidence (D) fairness (E) meticulousness

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. Chameleons, since they move quickly and adopt the color of their surroundings, are so difficult to ---- that even a careful observer can ---- their presence.

(A) eradicate. .notice (B) detect. .overlook (C) ignore. .misjudge (D) discern. .recognize (E) miss. .deduce

12. A ---- politician, Andrews worked hard and made so few mistakes that her opponents seemed to be ---- by contrast.

(A) shrewd. .sages (B) slothful. .drones (C) canny. .blunderers (D) dynamic. .firebrands (E) conscientious. .geniuses

13. Despite China's 2,000-year-old ---- medicinal plants, there was a period in Chinese history when the art of herbal medicine was almost ----.

(A) expertise in. .lost
(B) bewilderment over. .thwarted
(C) ignorance of. .eliminated
(D) affection for. .encouraged
(E) disapproval of. .censured

14. The author's tales of political chicanery might be taken as ---- were there not so much daily evidence of such tricks in this year's campaign.

(A) censorship (B) atonement (C) endorsement (D) mimicry (E) hyperbole

15. The committee warned that if its suggestions were not implemented, the problems would be ---- and eventually rendered insoluble.

(A) exacerbated (B) insurmountable (C) obliterated (D) vindicated (E) rebutted

GO ON TO THE NEXT PAGE

3

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. BIRD : MIGRATION :: (A) parrot : imitation
(B) ranger : conservation (C) bear : hibernation
(D) lawyer : accusation (E) traveler : location

17. RACQUET : TENNIS :: (A) springboard : diver
(B) horse : polo (C) glove : boxing
(D) club : golf (E) gun : hunting

18. OIL : LUBRICATE :: (A) adhesive : bond
(B) speedometer : accelerate (C) nail : hammer
(D) knife : sharpen (E) jacket : button

19. DESOLATE : INHABITANTS ::
(A) spacious : expanse (B) barren : vegetation
(C) shallow : hazards (D) residential : homes
(E) stagnant : permanence

20. HEAVY : WEIGHT :: (A) ancient : age
(B) warlike : force (C) brief : clarity
(D) bulky : strength (E) fertile : diversity

21. KERNEL : CORN :: (A) salt : pepper
(B) stem : flower (C) rind : orange
(D) grain : rice (E) trunk : tree

22. ANARCHY : ORDER ::
(A) ignorance : enlightenment
(B) skepticism : science
(C) hierarchy : statue
(D) autocracy : leader
(E) orthodoxy : tradition

23. PRESCRIPTIVE : RULE :: (A) imaginative : fact
(B) narrative : reader (C) symbolic : advice
(D) proverbial : adage (E) concrete : word

24. GIBE : SCORN :: (A) confess : punishment
(B) smile : awe (C) rebuff : friendship
(D) twitch : fury (E) chortle : exultation

25. DIGNITARY : AUGUST ::
(A) ingenue : sporting (B) rustic : abject
(C) zealot : solitary (D) parasite : modest
(E) penitent : contrite

GO ON TO THE NEXT PAGE

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

The resolution he had deliberately formed of not speaking to Sylvia on the subject of his love till he could announce to her parents the fact of his succession to Foster's business, and till he had patiently worked his way into her regard, was set aside during the present walk. He would speak to her of his passionate attachment before he left for an uncertain length of time and the certain distance of London. And all the modification on this point which his judgment could obtain from his impetuous and excited heart was that he would watch her words and manner well when he announced his approaching absence, and, if in them he read the slightest token of tender regretful feeling, he would pour out his love at her feet, not urging her to make any return or to express the feelings of which he hoped the germ was already budding in her. He would be patient with her; he could not be patient with himself. His heart beating, his busy mind rehearsing the probable coming scene, he turned onto the field-path that led to Haytersbank. Coming along it, and so meeting him, advanced Daniel Robson in earnest talk with Charley Kinraid. Kinraid, then, had been at the farm; Kinraid had been seeing Sylvia, her mother away. The thought of poor dead Annie Coulson flashed into Philip's mind. Could Kinraid be playing the same game with Sylvia? Philip set his teeth and tightened his lips at the thought of it.

26. The major subject of the passage is

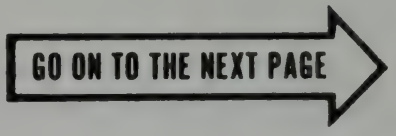
(A) Philip's recent success in business
(B) Kinraid's motives for seeing Sylvia
(C) Philip's thoughts and feelings about Sylvia
(D) Annie Coulson's role in Philip's life
(E) the nature of Sylvia's feelings for Philip

27. The passage implies which of the following about Sylvia?

(A) She is trying hard to conceal her budding feelings for Philip.
(B) She is likely to fall for a person like Charley Kinraid.
(C) She prefers Philip's overtures to those of Charley Kinraid.
(D) She has not expressed love for Philip.
(E) She is sure to regret Philip's imminent departure for London.

28. Which of the following best expresses the author's attitude toward Philip?

(A) Reluctant approval
(B) Detached sympathy
(C) Cold objectivity
(D) Veiled disdain
(E) Clear distaste



3

(This passage was written in 1963.)

Many recent advances in engineering–high-strength alloys and plastics no less than fluorescent lamps, transistors, and ferrites–are based on what we know about the structure of atoms. Even in the study of the phenomena of life, the stage has been reached where we examine effects of single atoms on living organisms. For these reasons alone it is clear that the study of atoms and their components is of great importance, but physicists have another strong incentive: curiosity. They just want to know what the world is made of, what the smallest particles are and how they behave. Therefore they are always ahead of practical applications; there always exists some knowledge about which people can ask, "What is it good for?" It is true that at the moment nobody can see any use for the "newer" particles--mesons and hyperons–yet in the past many new discoveries have, within a few decades, found some practical uses, or at least have become such indispensable parts of our knowledge that many practical advances would have been impossible without them. I do not foresee meson guns or hyperon boilers, but if applications for these particles are ever found, it is unlikely that even the most imaginative of present science-fiction writers will have envisaged them correctly.

29. The author mentions fluorescent lamps and transistors (lines 2-3) as examples of

(A) engineering devices that aid physicists in their research
(B) developments dependent on high-strength alloys
(C) inventions whose value was originally questioned by the public
(D) products of slightly less value than plastics and high-strength alloys
(E) applications derived from studies of atomic structure

30. The author apparently assumes that the question, "What is it good for?" (line 15), is usually asked in a tone of

(A) disdain (B) relief (C) enthusiasm (D) reverence (E) apprehension

31. The author's attitude toward physicists' curiosity is one of

(A) cautious skepticism
(B) apologetic embarrassment
(C) admiring support
(D) amused irritation
(E) reluctant tolerance

32. The passage as a whole is best described as a

(A) summary of recent advances derived from research on atomic particles
(B) challenge for physicists to find practical applications for the results of research
(C) prediction of future uses for the latest discoveries by atomic physicists
(D) defense of physicists' pursuit of knowledge about fundamental particles
(E) reminder to physicists that the world is impatient for new information about atoms

GO ON TO THE NEXT PAGE

As the manifestation of capitalism, the factory system led to a rise in the standard of living, to rapidly falling urban death rates and decreasing infant mortality, and to an unprecedented population explosion. In 1749, only one in four London children survived to a fifth birthday; by 1829, two out of three lived to a fifth birthday. England's population doubled from six million in 1750 to twelve million in 1820.

One is both morally unjust and historically inaccurate to blame capitalism for the condition of children during the Industrial Revolution; in fact, capitalism brought enormous improvement over their condition in the preceding age. The sources of the inaccurate perceptions of capitalism were ill-informed, emotional novelists and poets like Dickens and Mrs. Browning; fanciful medievalists like Southey; political tract writers like Engels and Marx, posturing as economic historians. All painted vague, rosy pictures of a lost "golden age" of the working classes, an age that they claimed was destroyed by the Industrial Revolution. Historians have not supported their assertions. Investigation and common sense have deglamorized prefactory domestic industry. In that system, the worker made a costly initial investment in, or paid heavy rentals for, a loom or frame; thus, the worker had to bear the speculative risks involved. His diet was drab and meager, and he was forced to enlist the labor of his wife and children. There was nothing romantic or enviable about a family living and working together in a badly lighted, improperly ventilated, poorly constructed cottage.

33. The author includes statistical information specifically to
    (A) dispute the cause-and-effect relationship between the development of the factory system and the growth in population
    (B) support the claim that life in preindustrial England was worse than life in the industrial age
    (C) emphasize the unfortunate plight of children in the 1800's
    (D) prove that Engels and Marx were not economic historians
    (E) demonstrate that the Industrial Revolution had ended in England by 1829

34. It can be inferred from the passage that the author's attitude toward the factory system is one of
    (A) dismay at the way it separated workers from their families
    (B) fear of the rapid population growth it spawned
    (C) suspicion of the social and economic changes it initiated
    (D) enthusiasm about the way it inspired writers to produce great works
    (E) support of the improvement it brought to the lives of workers

35. Which of the following best describes the author's apparent purpose in this passage?
    (A) To correct a distorted picture of capitalism's early effects
    (B) To help workers understand the history of the labor movement
    (C) To demonstrate the impact of modern statistics on historical research
    (D) To explain how much capitalism has changed over the years
    (E) To create a sense of outrage over the injustices of the early days of capitalism

GO ON TO THE NEXT PAGE

3

The seventeenth century was the golden age of the amateur. There was not yet a recognition of the fact that there are different categories of knowledge and phenomena. In the desire to solve the great puzzle of the world and to discover relations between groups of phenomena, the love of tracing analogy and the temptation to press an analogy once discerned ever further and further had led many thinkers of the Middle Ages to false conclusions in many matters. The importance of the proper use of analogy and its indispensable service to scientific thought are, of course, not to be questioned. But by the extensive misuse of analogy, and by often reasoning logically from false premises, thinkers had managed to construct a scheme of the universe which was fairly coherent but which, when tested by observation and experiment, was bound to crumble. The seventeenth century inherited such schemes, and much of the ancient and medieval misinformation about natural phenomena remained part of the mental climate of Europe until well on in the century. The important fact, however, that emerges from a study of the scientific thought and opinion of the century is not that such and such persons believed such and such things, but that a good many persons were beginning to question old beliefs, and to discover that much that had been accepted on the authority of the ancients was not true.

Philosophy tended more and more during the century to concern itself with "reason" and with matters amenable to scientific experimentation and mathematical demonstration. The "laws of nature" occupied an increasingly larger share of human thought, and began to take their place by the side of the "revealed truth" of the Bible. To the more devout scientists these studies in natural philosophy seemed to supplement and re-enforce rather than to supersede the truths of the Bible. Sir Thomas Browne, an ardent believer in the Bible, commended those who magnify God by "judicious inquiry into His acts and deliberate researches into His creatures." By the end of the century, thanks to the work of the scientists and philosophers, a state of mind had been induced in which such phenomena as plagues, conflagrations, and death came to be viewed less as evidence of divine displeasure than as natural effects and circumstances, and natural methods were undertaken to deal with them. Before the century was over, both fire insurance and life insurance had arrived in England.

36. The author suggests that philosophers of the Middle Ages were most concerned with

(A) discovering general truths about the universe
(B) proving revealed truths scientifically
(C) examining individual natural phenomena
(D) eliminating misconceptions about phenomena
(E) combining scientific knowledge with philosophy

37. The author mentions the introduction of fire insurance and life insurance in England (lines 50-51) as

(A) an illustration of the effects of the later seventeenth-century view of natural phenomena
(B) a refutation of Sir Thomas Browne's view of scientific study
(C) an example of the influence of economic growth on trends of thought
(D) an example of the seventeenth-century recognition of the errors of medieval thought
(E) an illustration of the seventeenth-century interest in the classification of phenomena

38. The author attributes the misconceptions that seventeenth-century thinkers inherited from the Middle Ages most directly to the medieval

(A) attempts to apply science to religious ideas
(B) failure to recognize different categories of knowledge and phenomena
(C) tendency to carry analogies to extremes
(D) reliance on the incorrect interpretations of their predecessors
(E) failure to use logical reasoning in constructing their theories

39. The author quotes Sir Thomas Browne (lines 41-43) in order to

(A) exemplify the mistaken ideas carried over from the Middle Ages into the seventeenth century
(B) emphasize the increased importance of religious teachings in seventeenth-century thought
(C) illustrate the basic differences between medieval and seventeenth-century ideas
(D) suggest that seventeenth-century studies of natural phenomena were approached cautiously
(E) demonstrate that scientific and revealed truth were not universally regarded as incompatible in the seventeenth century

40. The author is primarily concerned with

(A) making a distinction between scientific thought and scientific opinion in the seventeenth century
(B) describing the decline of religion as an influence on philosophy in the seventeenth century
(C) discussing the mental climate of Europe in the seventeenth century
(D) outlining the religious philosophy of seventeenth-century Europe
(E) discussing medieval misconceptions that influenced seventeenth-century thought

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 5
Time—30 minutes
35 QUESTIONS

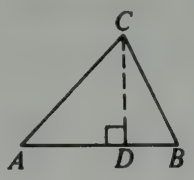
In this section solve each problem, using any available space on the page for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.
If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. If $N \times \frac{5}{14} = \frac{5}{14} \times \frac{7}{9}$, then $N =$

   (A) $\frac{7}{9}$ (B) $\frac{9}{7}$ (C) 5 (D) 7 (E) 14

2. Which of the following numbers is NOT the product of 11 and an integer?

   (A) 11,111 (B) 111,111 (C) 11,111,111
   (D) 1,111,111,111 (E) 111,111,111,111

3. If the area of a square is $16y^2$, what is the length of a side in terms of y?

   (A) y (B) $2y$ (C) $4y$ (D) y^2 (E) $4y^2$

4. If $12p - 9q = 4$, then $4p - 3q =$

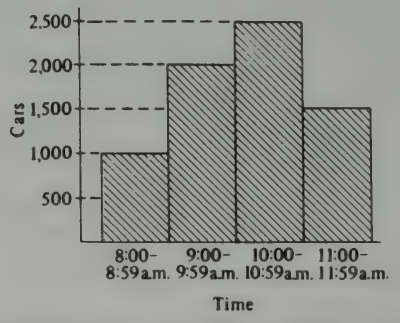
   (A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{3}{4}$ (D) $\frac{4}{3}$ (E) 3

5. The positive difference between the cubes of any two consecutive positive integers is always

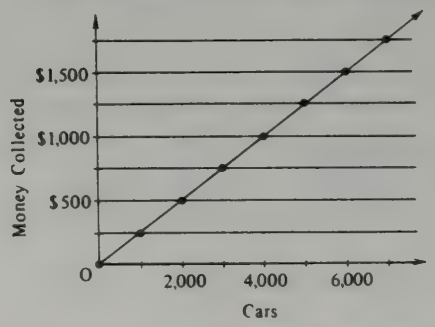
   (A) an even integer
   (B) an odd integer
   (C) the product of 3 and an integer
   (D) the square of an integer
   (E) the cube of an integer

Questions 6-7 refer to the following graphs.

THE NUMBER OF CARS PER HOUR PASSING THROUGH TOLLBOOTH X BETWEEN 8:00 A.M. AND 11:59 A.M.



MONEY COLLECTED IN TOLLBOOTH X VS. THE NUMBER OF CARS PASSING THROUGH IT



6. The number of cars passing through tollbooth X between 9:00 a.m. and 11:59 a.m., inclusive, is

   (A) 3,500 (B) 4,000 (C) 5,500
   (D) 6,000 (E) 7,000

7. Of the following, which is the earliest time at which $750 in tolls could have been collected?

   (A) 8:00 a.m. (B) 8:59 a.m. (C) 9:59 a.m.
   (D) 10:59 a.m. (E) 11:59 a.m.

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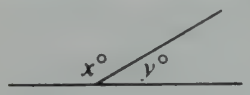
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Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet blacken space

A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

Notes: 1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. In a given question, a symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ●

	Column A	Column B
8.	$\left(\frac{1}{3}\right)^2$	$\left(\frac{1}{2}\right)^2$
9.	The number of hours in a week	150
10.	$\frac{1}{2} + \frac{1}{3} + \frac{1}{11}$	$\frac{1}{11} + \frac{1}{2} + \frac{1}{4}$
11.	Number of sides of a triangle	Number of sides of a quadrilateral

$x > z$
$y < z$

	Column A	Column B
12.	z	0
13.	The average (arithmetic mean) of 1, 2, and k	The average (arithmetic mean) of 2, 3, and k

	Column A	Column B

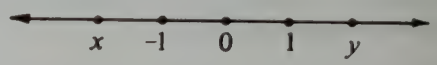
$T = \{3, 4, 6, 8, 9, 12\}$

A number is selected at random from set T above.

14.	The probability that the number selected will be a multiple of 3	The probability that the number selected will be a multiple of 4

$x + y = 3$
$y < \frac{3}{2}$

15.	x	y



16.	$x - y$	$y - x$

If John had 30 more pencils, he would have twice as many as he actually has. If Carl had 20 fewer pencils, he would have half as many as he actually has.

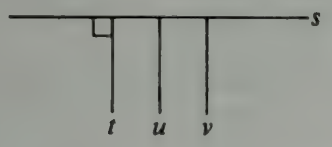
17.	The number of pencils John actually has	The number of pencils Carl actually has

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SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

Answer: **A** if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

	Column A	Column B



s, t, u, and v are lettered line segments.

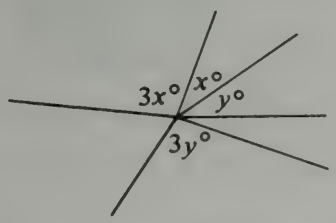
$t \parallel u \parallel v$

18. The number of pairs of parallel lettered line segments | The number of pairs of perpendicular lettered line segments

---

$\frac{1}{N} > 1$

19. N^2 | 1



Note: Figure not drawn to scale.

Six line segments meet at a point.

20. x | y

$x + y - z = 10$

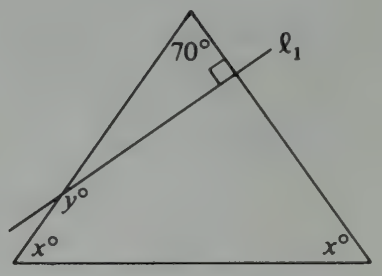
$x + y + z = 5$

21. $2x + 2y$ | 14

$xy \neq 0$

22. x^2y | $-xy^2$

	Column A	Column B

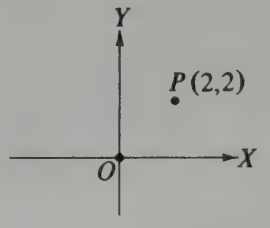


23. $2x$ | y

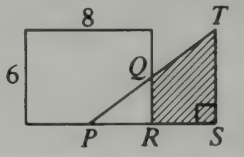
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a is a positive whole number.

24. $a(a^a)$ | $(a + 1)^a$



25. Area of a circle with center at point P and circumference passing through O | Area of a circle with circumference passing through points O and P



P and Q are midpoints on two sides of the rectangle and Q and R are midpoints on two sides of $\triangle PST$.

26. Perimeter of the shaded region | 18

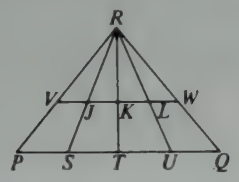
$5 < xy < 20$ and x and y are integers.

27. xy | $x + y$

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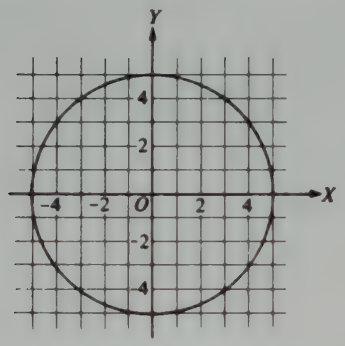
Solve each of the remaining problems in this section using any available space for scratchwork. Then decide which is the best of the choices given and blacken the corresponding space on the answer sheet.



28. In $\triangle PRQ$ above, $VW \parallel PQ$, and S, T, and U divide PQ into four equal parts. What is the ratio $\frac{\text{length of } JL}{\text{length of } VW}$?

(A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$

(E) It cannot be determined from the information given.



29. What is the circumference of the circle in the figure above?

(A) 5π (B) 10π (C) $\frac{25}{2}\pi$

(D) 20π (E) 25π

---

30. If m and n are positive integers and $m < n$, which of the following must be greater than $\frac{m}{n}$?

I. $\frac{m+1}{n+1}$ II. $\frac{m+1}{n}$ III. $\frac{m}{n+1}$

(A) I only (B) II only (C) III only

(D) I and II only (E) II and III only

31. A machine produces a thingamajig every 0.025 seconds. How many thingamajigs does it produce in an hour?

(A) 2,400 (B) 9,000 (C) 12,000

(D) 14,400 (E) 144,000

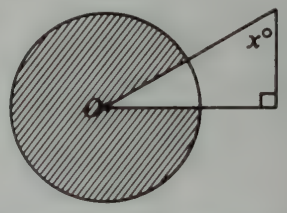
32. If A, B, and C are points on a line, in that order, and $AC - AB = 4$, what is the ratio $\frac{AB}{AC}$?

(A) 1 to 5
(B) 1 to 4
(C) 1 to 3
(D) 1 to 2
(E) It cannot be determined from the information given.

33. In a class of 80 seniors, there are 3 boys for every 5 girls. In the junior class there are 3 boys for every 2 girls. If the two classes combined have an equal number of boys and girls, how many students are in the junior class?

(A) 200 (B) 192 (C) 120

(D) 100 (E) 32



Note: Figure not drawn to scale.

34. The circle shown above has a center O and radius of length 5. If the area of the shaded region is 20π, what is the value of x?

(A) 18 (B) 36 (C) 45 (D) 54 (E) 72

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35. The price of 10 pounds of apples is d dollars. If the apples weigh an average of 1 pound for every 6 apples, which of the following is the average price, in cents, of a dozen such apples?

(A) $20d$ (B) $\frac{50d}{3}$ (C) $5d$ (D) $\frac{5d}{3}$ (E) $\frac{d}{20}$

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

Correct Answers for Scholastic Aptitude Test
Form Code 1F

VERBAL		MATHEMATICAL	
Section 1	**Section 3**	**Section 2**	**Section 5**
1. C	1. D	1. B	1. A
2. E	2. E	2. E	2. A
3. A	3. B	3. C	3. C
4. C	4. C	4. D	4. D
5. E	5. B	5. C	5. B
6. B	6. D	6. D	6. D
7. D	7. C	7. A	7. C
8. A	8. D	8. C	*8. B
9. D	9. B	9. E	*9. A
10. D	10. C	10. E	*10. A
11. E	11. B	11. B	*11. B
12. E	12. C	12. C	*12. D
13. C	13. A	13. B	*13. B
14. C	14. E	14. E	*14. A
15. E	15. A	15. C	*15. A
16. D	16. C	16. E	*16. B
17. B	17. D	17. C	*17. B
18. E	18. A	18. C	*18. C
19. D	19. B	19. B	*19. B
20. B	20. A	20. A	*20. D
21. A	21. D	21. E	*21. A
22. D	22. A	22. A	*22. D
23. A	23. D	23. A	*23. B
24. C	24. E	24. C	*24. D
25. C	25. E	25. C	*25. D
26. B	26. C		*26. C
27. C	27. D		*27. D
28. B	28. B		28. B
29. D	29. E		29. B
30. B	30. A		30. D
31. D	31. C		31. E
32. B	32. D		32. E
33. A	33. B		33. D
34. B	34. E		34. A
35. E	35. A		35. A
36. B	36. A		
37. E	37. A		
38. A	38. C		
39. A	39. E		
40. D	40. C		
41. C			
42. A			
43. A			
44. E			
45. C			

*Indicates four-choice questions. (All of the other questions are five-choice.)

The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. The SAT-verbal test has 85 questions with five answer choices each. If, for example, a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

  $$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

  Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.
- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty among different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing SAT scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

How to Score the Test

You can verify the College Board SAT scores reported to you recently by using the information in this booklet along with the copy of your answer sheet. *Before you begin, check that the first two characters (number and letter) of the form code you marked in item 3 on your answer sheet are the same as the form code printed on the front of this booklet.* Compare the responses shown on the copy of your answer sheet with the list of correct answers.

SAT-Verbal Sections 1 and 3

Step A: Count the number of correct answers for *section 1* and record the number in the space provided on the worksheet on the next page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for *section 3* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals. Enter the resulting figure on the worksheet.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step E: To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on page 148. Enter this figure on the worksheet.

SAT-Mathematical Sections 2 and 5

Step A: Count the number of correct answers and the number of incorrect answers for *section 2* and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal C}$$

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals. Enter the resulting figure on the worksheet.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step F: To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on page 148. Enter this figure on the worksheet.

SAT SCORING WORKSHEET

FORM CODE 1F

SAT-Verbal Sections

A. Section 1: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal A

B. Section 3: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. SAT-verbal reported scaled score (See the conversion table on the back cover.) [ ] SAT-verbal score

SAT-Mathematical Sections

A. Section 2: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal A

B. Section 5: Questions 1 through 7 and 28 through 35 (5-choice) _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal B

C. Section 5: Questions 8 through 27 (4-choice) _____ (no. correct) − ⅓ (_____ no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. SAT-mathematical reported scaled score (See the conversion table on the back cover.) [ ] SAT-math score

Score Conversion Table
Scholastic Aptitude Test
Form Code 1F

Raw Score	College Board Reported Score		Raw Score	College Board Reported Score	
	SAT-Verbal	SAT-Math		SAT-Verbal	SAT-Math
85	800		40	460	630
84	790		39	450	620
83	780		38	440	610
82	770		37	440	600
81	760		36	430	590
80	750		35	420	580
79	740		34	420	570
78	730		33	410	560
77	720		32	400	560
76	710		31	400	550
75	700		30	390	540
74	690		29	380	530
73	680		28	380	520
72	670		27	370	510
71	660		26	360	500
70	660		25	360	490
69	650		24	350	490
68	640		23	350	480
67	640		22	340	470
66	630		21	330	460
65	620		20	330	450
64	620		19	320	440
63	610		18	310	430
62	600		17	310	420
61	600		16	300	420
60	590	800	15	290	410
59	580	790	14	290	400
58	580	780	13	280	390
57	570	780	12	270	380
56	560	770	11	270	370
55	560	760	10	260	360
54	550	750	9	250	350
53	540	740	8	250	350
52	540	730	7	240	340
51	530	720	6	230	330
50	520	710	5	230	320
49	520	710	4	220	310
48	510	700	3	210	300
47	500	690	2	210	290
46	500	680	1	200	280
45	490	670	0	200	270
44	480	660	– 1	200	270
43	480	650	– 2	200	260
42	470	640	– 3	200	250
41	460	640	– 4	200	240
			– 5	200	230
			– 6	200	220
			– 7	200	210
			– 8 or below	200	200